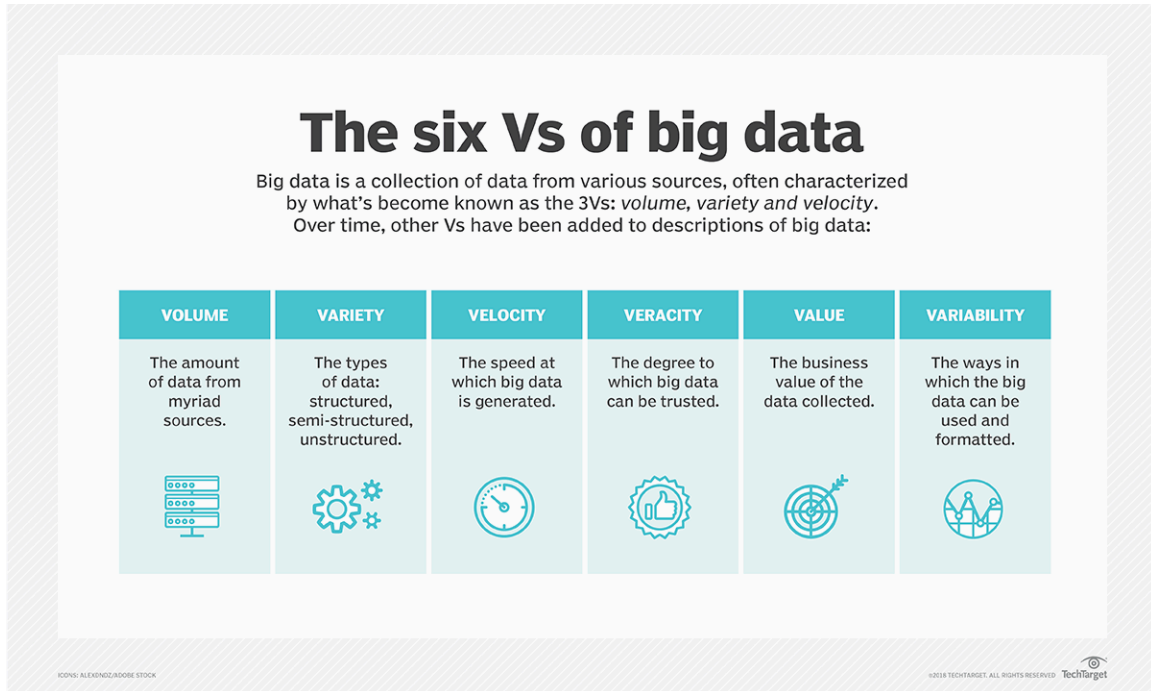


EL10: Big data

Big data



Biggish data

My definition:

- Only tabular data
- Original data might fit into physical RAM but ...
- might be multiplied due to temporary necessity
- physical RAM might be constrained due to other processes/settings
- things might just take too much time ... and life is short ...

Software

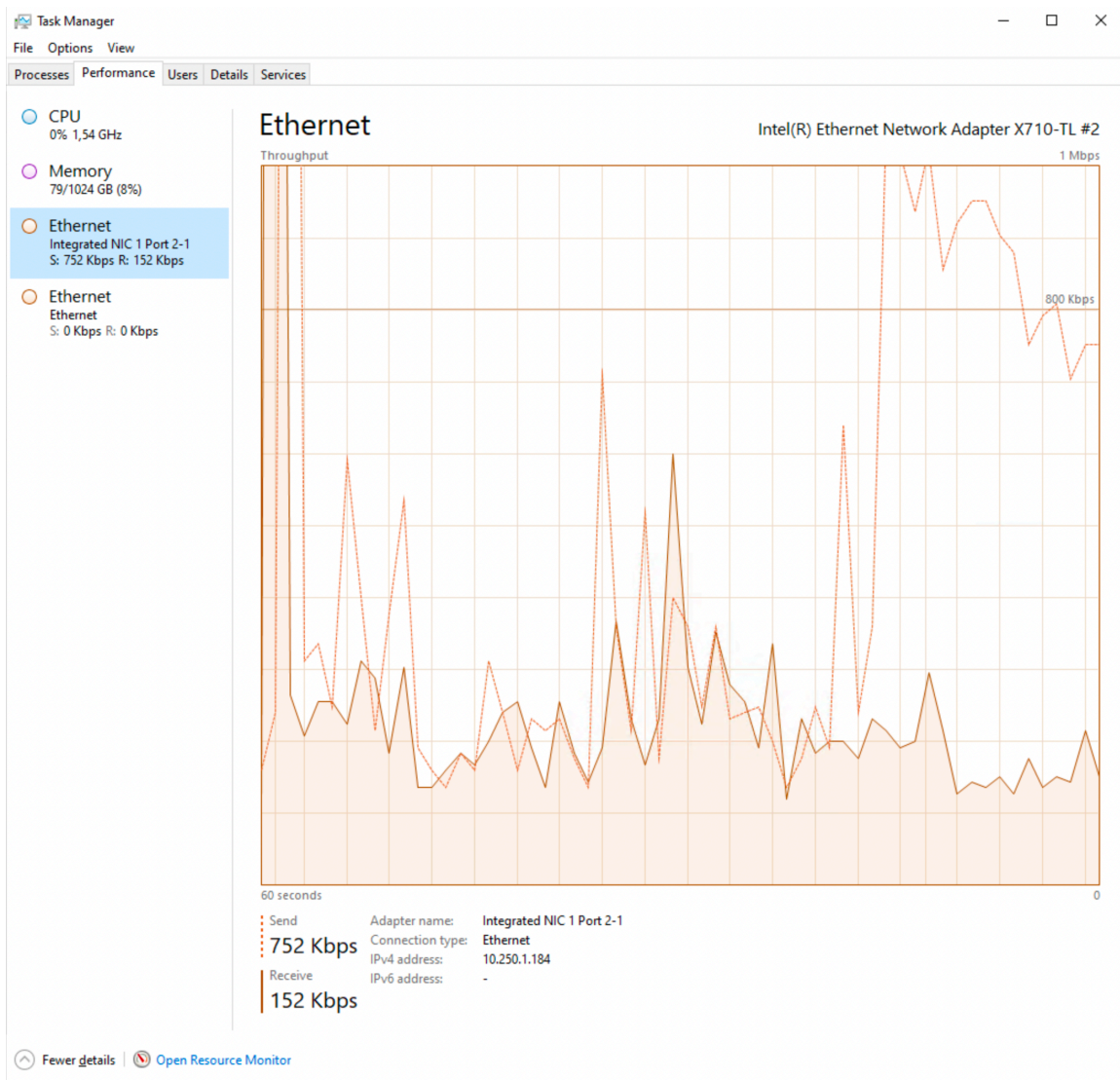
- SAS for big data files from register holders
- R for everything else
- Database?

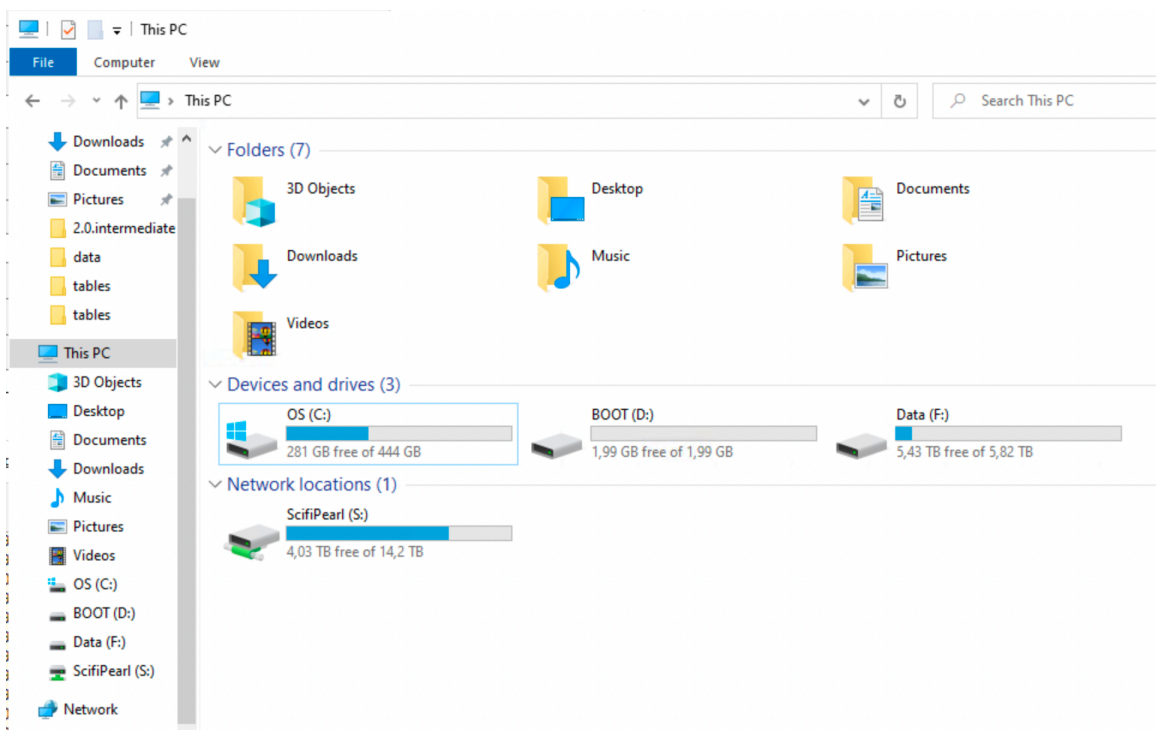
File format

File storage

- Avoid reading big files over ethernet connection

- data stored on network drive
 - Usually faster to first copy file to local SSD drive
 - `fs::file_copy(path, new_path)`
 - (“*libuv provides a wide variety of cross-platform sync and async file system operations.*”)
 - obviously only if “local drive” is also in the (same) secure environment
 - work with the project on disk
 - move back
 - Although not always possible with very large files it seems
-





Comparison vs base equivalents

fs functions smooth over some of the idiosyncrasies of file handling with base R functions:

- Vectorization. All **fs** functions are vectorized, accepting multiple paths as input. Base functions are inconsistently vectorized.
- Predictable return values that always convey a path. All **fs** functions return a character vector of paths, a named integer or a logical vector, where the names give the paths. Base return values are more varied: they are often logical or contain error codes which require downstream processing.
- Explicit failure. If **fs** operations fail, they throw an error. Base functions tend to generate a warning and a system dependent error code. This makes it easy to miss a failure.
- UTF-8 all the things. **fs** functions always convert input paths to UTF-8 and return results as UTF-8. This gives you path encoding consistency across OSes. Base functions rely on the native system encoding.
- Naming convention. **fs** functions use a consistent naming convention. Because base R's functions were gradually added over time there are a number of different conventions used (e.g. `path.expand()` vs `normalizePath()`; `Sys.chmod()` vs `file.access()`).

sync_project()

- I have a function `sync_project()` in my `.Rprofile`
 - `usethis::edit_r_profile()`
- However, I don't include all data files in each individual project
- "Chek out" all files from network storage to disk
- Work with it
- Sync back to allow back-ups and common folder structure
- Git-repo kept for the network storage

```
sync_project <- function (from, to)
{
  loc_dirs <- c("S:/Projects/", "F:/xbuler/")
  loc_dirs_exp <- paste(loc_dirs, collapse = "|")
  proj <- gsub(loc_dirs_exp, "", getwd())
  locations <- paste0(loc_dirs, proj)
  locations_exp <- paste(paste0(locations, "/"), collapse = "|")
  stopifnot(getwd() %in% locations)
```



```

from <- if (missing(from))
  getwd()
to <- if (missing(to))
  setdiff(locations, from)
ans <- askYesNo(paste("Sync from:\n ", from, "\n To: \n",
  to))
stopifnot(ans)
if (!fs::dir_exists(to)) {
  fs::dir_create(to)
  message("Target folder did not exist but has been created!")
}
fr <- dplyr::mutate(dplyr::select(fs::dir_info(from, all = TRUE,
  recurse = TRUE, regexp = ".git", invert = TRUE), path,
  type, size, modification_time), file = gsub(locations_exp,
  "", path), path_to = paste(to, file, sep = "/"))
t <- dplyr::mutate(dplyr::select(fs::dir_info(to, all = TRUE,
  recurse = TRUE), path, type, size, modification_time),
  file = gsub(locations_exp, "", path))
new <- dplyr::anti_join(fr, t, "file")
new_dirs <- dplyr::filter(new, type == "directory")
if (nrow(new_dirs)) {
  print(dplyr::select(new_dirs, file), n = Inf)
  ans <- askYesNo(paste(nrow(new_dirs), " dir(s) not in 'to'. Should I
create these?"))
  if (ans)
    fs::dir_create(new_dirs$path_to)
}
new_files <- dplyr::filter(new, type == "file")
if (nrow(new_files)) {
  print(dplyr::select(new_files, file), n = Inf)
  ans <- askYesNo(paste(nrow(new_files), " file(s) not in 'to'. Should I
copy them?"))
  if (ans)
    purrr::walk2(new_files$path, new_files$path_to, fs::file_copy,
      .progress = TRUE)
}
diff_mod <- dplyr::filter(dplyr::left_join(fr, t, "file",
  suffix = c("_from", "_to")), type_from == "file",
modification_time_to !=
  modification_time_from)
if (nrow(diff_mod)) {
  print(dplyr::select(diff_mod, file,
tidyselect::matches("modification_time|size")),
    n = Inf)
  ans <- askYesNo(paste(nrow(diff_mod), " file(s) have been updated.
Should I replace the older version?"))
  if (ans) {
    purrr::walk2(diff_mod$file, diff_mod$path_to, fs::file_copy,

```

```

        overwrite = TRUE, .progress = TRUE)
    }
  }
  dlt <- dplyr::filter(dplyr::anti_join(t, fr, "file"), !startsWith(file,
    ".git"))
  if (nrow(dlt)) {
    print(dlt, n = Inf)
    ans <- askYesNo(paste(nrow(dlt), " file(s) have been deleted in
'from'. Should I delete them in 'to'?"))
    if (ans) {
      purrr::walk(dlt$path, fs::file_delete, .progress = TRUE)
    }
  }
}

```

👉 Reading SAS files in R

- Common to get large register files in SAS-format (SoS, SCB)
- 👍 SAS can handle that (files on disk) 😊
- 👉 I am a terrible SAS user 😞
- 👉 Seems to be no easy way to read only parts of SAS-files into R directly
- `haven::read_sas()`?

`haven::read_sas()`

- 👍 May work if big enough RAM ...
- 👉 Whole file is first read into RAM (`readr::datasource(data_file)`)
- 👉 Most time spent on `df_parse_sas_file()`
- (Same for .dta-files [Stata])

```
haven::read_sas
```

```

function (data_file, catalog_file = NULL, encoding = NULL, catalog_encoding =
encoding,
  col_select = NULL, skip = 0L, n_max = Inf, cols_only = deprecated(),
  .name_repair = "unique")
{
  if (lifecycle::is_present(cols_only)) {
    lifecycle::deprecate_warn("2.2.0", "read_sas(cols_only)",
      "read_sas(col_select)")
    stopifnot(is.character(cols_only))
    col_select <- quo(c(!!!cols_only))
  }
  else {
    col_select <- enquos(col_select)
  }
}

```

```

}
if (is.null(encoding)) {
  encoding <- ""
}
cols_skip <- skip_cols(read_sas, !!col_select, data_file,
  encoding = encoding)
n_max <- validate_n_max(n_max)
spec_data <- readr::datasource(data_file)
if (is.null(catalog_file)) {
  spec_cat <- list()
}
else {
  spec_cat <- readr::datasource(catalog_file)
}
switch(class(spec_data)[1], source_file = df_parse_sas_file(spec_data,
  spec_cat, encoding = encoding, catalog_encoding = catalog_encoding,
  cols_skip = cols_skip, n_max = n_max, rows_skip = skip,
  name_repair = .name_repair), source_raw = df_parse_sas_raw(spec_data,
  spec_cat, encoding = encoding, catalog_encoding = catalog_encoding,
  cols_skip = cols_skip, n_max = n_max, rows_skip = skip,
  name_repair = .name_repair), cli_abort("This kind of input is not
handled."))
}
<bytecode: 0x1404bfad0>
<environment: namespace:haven>

```

👍 Export from SAS to csv

- I use SAS code to export only necessary data to csv
- I might use a prefiltering list exported from R to limit the data
- SAS Viya more flexible (I've heard) but not SAS 9.4
- Use (set=1000) etc for tests and then run over night
- csv-files exported to local SSD drive (and then synced back to storage folder)

ChatGPT

Jag har exporterat en **csv-fil från R** som innehåller en lista med **löpnummer** på individer. Jag vill nu använda **SAS**. Först vill jag läsa in CSV-filen. Sedan vill jag **filtrera** ett dataset i SAS (vi kan kalla det pelle) som har en kolumn med samma löpnummer som i CSV-filen (jag vill bara ha med de fall där det finns en match). Därefter vill jag **exportera** resultatet till en **ny csv-fil**. Jag vill att detta ska ske på ett **effektivt och snabbt** sätt. Hjälp med **SAS-syntax** för detta

```
/* Läs in CSV-filen i SAS */
```

```

/* Läs in filen med löpnummer till ett SAS-dataset (vi kallar det `individer`)
*/
proc import datafile="din_sökväg_till_csv_filen.csv"
    out=individer>
    dbms=csv
    replace;
    getnames=yes; /* Sätt till no om csv-filen saknar rubriker */
run;

/* Om CSV-filen saknar rubrik kan du lägga till följande steg för att namnge
kolumnen manuellt */
data individer;
    set individer(rename=(var1=löpnummer)); /* Byt "var1" till korrekt
variabelnamn */
run;

/* Filtrera datasetet `pelle` baserat på löpnummer */

/* Vi använder en hash-lookup för att göra filtreringen effektiv. Hash-lookup
är snabbare än traditionella merge-metoder. */
data pelle_filtered;
    if _n_ = 1 then do;
        declare hash h(dataset: "individer");
        h.definekey("löpnummer");
        h.definedone();
    end;
    set pelle;
    if h.find() = 0; /* Endast matchande löpnummer */
run;

/* Exportera det filtrerade datasetet till en ny CSV-fil */

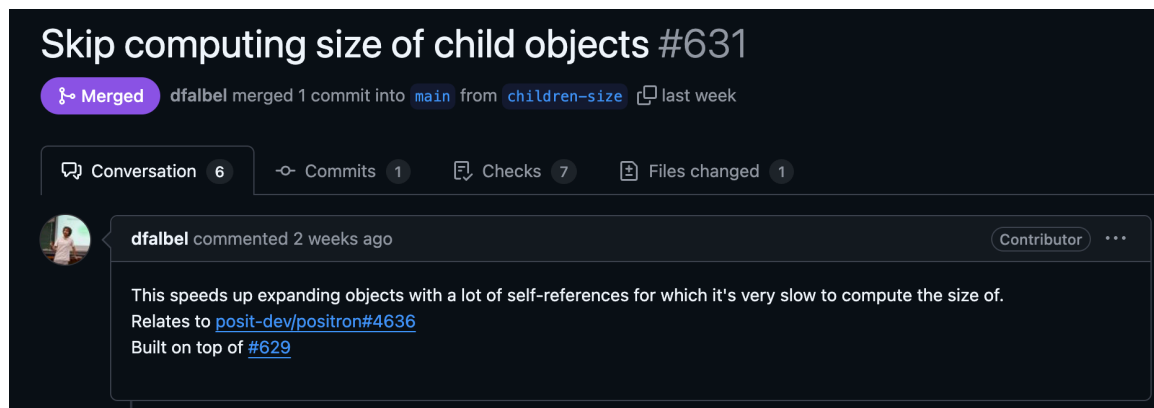
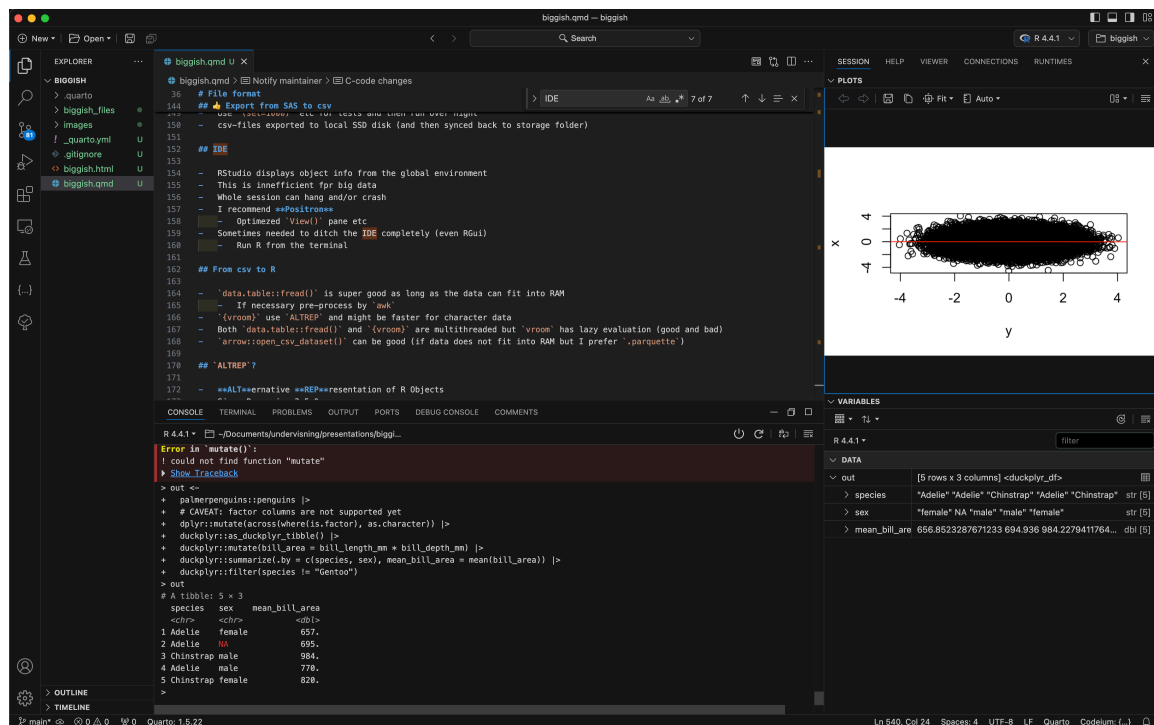
/* Använd proc export för att spara resultatet */
proc export data=pelle_filtered
    outfile="din_sökväg_till_ny_csv_fil.csv"
    dbms=csv
    replace;
run;

/* Effektivitetsoptimering:
Om både `individer` och `pelle` är mycket stora,
kan du sortera båda och använda data step merge
istället för hash-lookup. Hash-metoden är dock
generellt snabbare om det är många rader i båda
tabellerna. */

```

IDE

- RStudio displays object info from the global environment
- This is inefficient for big data
- Whole session can hang and/or crash
- I recommend [Positron](#)
 - Optimized View() pane etc
 - Active and enthusiastic development
- Sometimes needed to ditch the IDE completely (even RGui)
 - Run R from the terminal



```
Rterm
C:\Windows\system32>"C:\Program Files\R\R-4.4.1\bin\R.exe"

R version 4.4.1 (2024-06-14 ucrt) -- "Race for Your Life"
Copyright (C) 2024 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

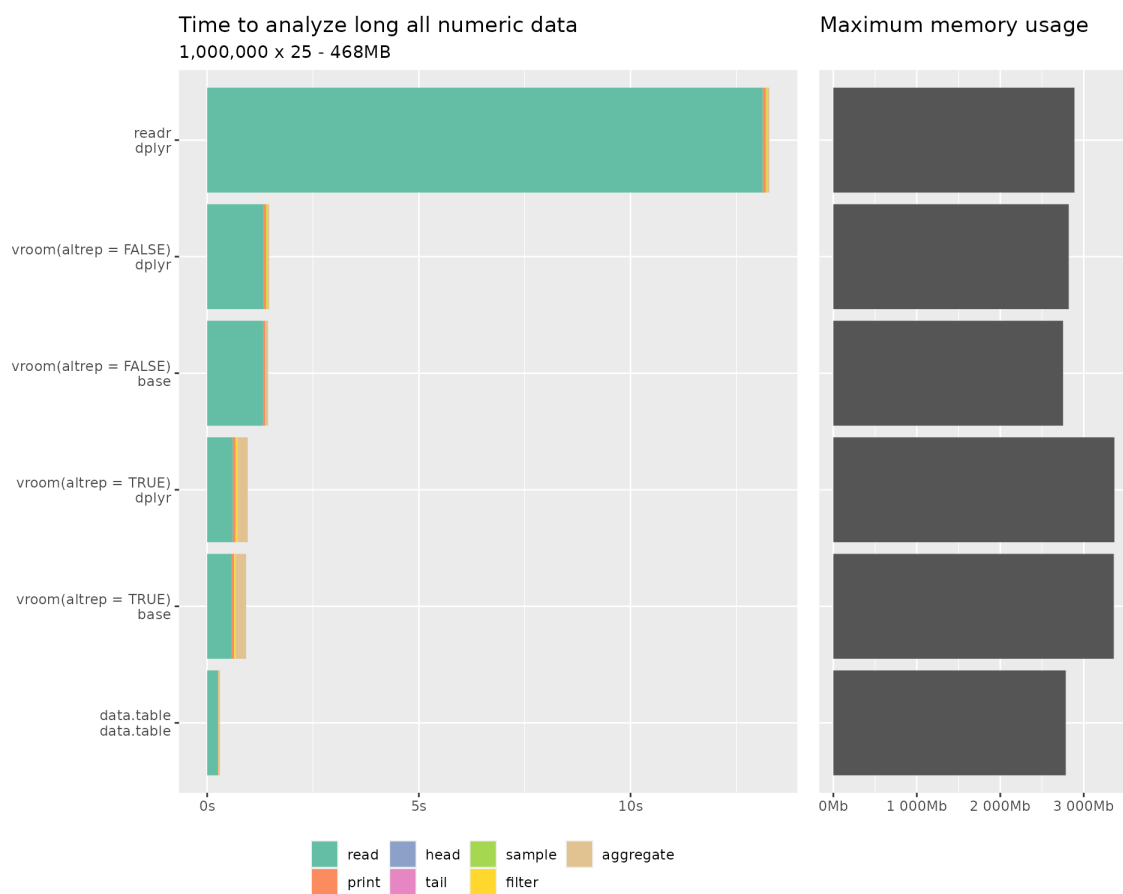
R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

Loading required package: digest
Loading required package: tibble
> _
```

From csv to R

- `data.table::fread()` is super good as long as the data can fit into RAM
 - If necessary pre-process by `awk`
 - `{vroom}` use ALTREP and might be faster for character data
 - Both `data.table::fread()` and `{vroom}` are multithreaded but `vroom` has lazy evaluation (good and bad)
 - `arrow::open_csv_dataset()` can be good (if data does not fit into RAM but I prefer `.parquet` if so)
-




```
man
General Commands Manual
AWK(1)

NAME
    awk - pattern-directed scanning and processing language

SYNOPSIS
    awk [ -F fs ] [ -v var=value ] [ 'prog' | -f progfile ] [ file ... ]

DESCRIPTION
    Awk scans each input file for lines that match any of a set of patterns specified literally in prog or in one or more files specified as -f progfile. With each pattern there can be an associated action that will be performed when a line of a file matches the pattern. Each line is matched against the pattern portion of every pattern-action statement; the associated action is performed for each matched pattern. The file name - means the standard input. Any file of the form var=value is treated as an assignment, not a filename, and is executed at the time it would have been opened if it were a filename. The option -v followed by var=value is an assignment to be done before prog is executed; any number of -v options may be present. The -F fs option defines the input field separator to be the regular expression fs.

    An input line is normally made up of fields separated by white space, or by the regular expression FS. The fields are denoted $1, $2, ..., while $0 refers to the entire line. If FS is null, the input line is split into one field per character.

    A pattern-action statement has the form:
        pattern { action }

    A missing { action } means print the line; a missing pattern always matches. Pattern-action statements are separated by newlines or semicolons.

    An action is a sequence of statements. A statement can be one of the following:

        if( expression ) statement [ else statement ]
        while( expression ) statement
        for( expression ; expression ; expression ) statement
        for( var in array ) statement
        do statement while( expression )
        break
        continue
        { [ statement ... ] }
        expression # commonly var = expression
        print [ expression-list ] [ > expression ]
        printf format [ , expression-list ] [ > expression ]
        return [ expression ]
        next # skip remaining patterns on this input line
        nextfile # skip rest of this file, open next, start at top
        delete array[ expression ] # delete an array element
        delete array # delete all elements of array
```

ALTREP?

- **AL**Ternative **RE**Presentation of R Objects
- Since R version 3.5.0
- Applies to the C-level representation of R objects under the hood
- Used by some base R packages/functions + specialized packages

allow vector data to be in a memory-mapped file; distributed, e.g. within Apache Spark or Hadoop; shared with other applications, e.g. with Apache Arrow; allow compact representation of arithmetic sequences; allow adding meta-data to objects; allow computations/allocations to be deferred; support alternative representations of environments.

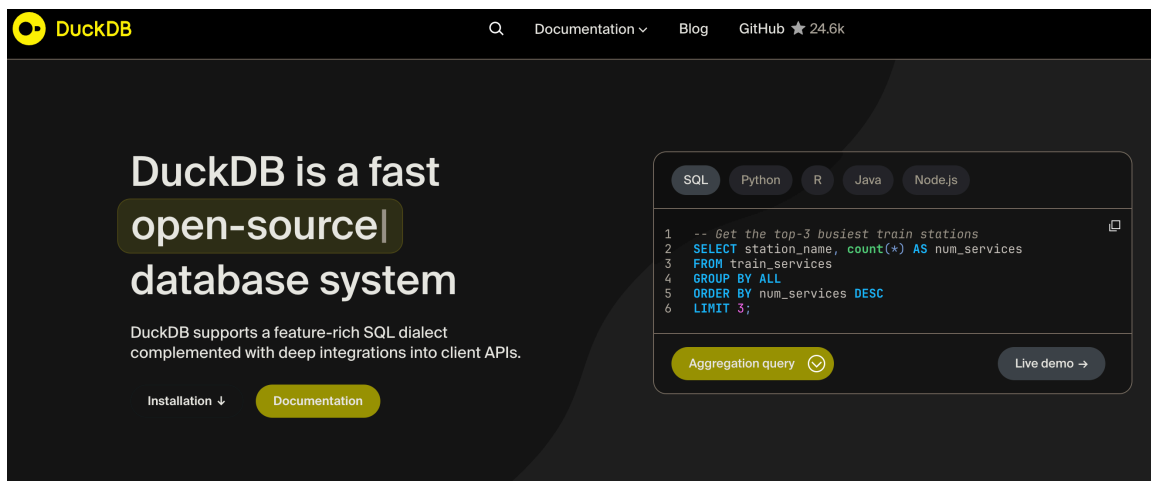
Database?

Database?

- I don't want to administer Microsoft SQL Server myself!
 - (They had R support for a couple of years but not any more)
- SQLite is easy to use ...
- but (as SQL server) it is row based 😞
- Some complicated ones are column based 😊 (Snowflake, Google Big Query)
 - ... but I don't want to administer those either ...
- DuckDB comes for rescue!

! Indices

Never forget to set proper indices if needed!



DuckDB

- Free and open source (backed by foundation)
- Column based
- Very easy to set up
- Very efficient!
- Integration with tidyverse in R
- Can be used both with disk data and data in RAM
- ... but I had some bad experience and now I am scared 😬

From CSV to Parquet

- CSV is not very nice 😞
 - No meta data
 - Sensitive to errors (missing values, decimal point, numbers with leading 0 etc, although fread can take care of this)
 - Encoding, BOM, whatever ...
- Write to Parquet files
- I don't think it is possible to convert csv to parquet on disk (first read csv, then write parquet)
 - But it can be done in chunks (read portion of csv and write to individual parquet file which are later "combined")



Parquet

- From Apache Arrow
- Open format
- flat files with possible hierarchical structure by transparent structure of folders and filenames
- Integrates with DuckDB (which integrates with R/tidyverse)
- See example later!

R specific data files?

- I use .qd-files (previously .qs) to cache non-tabular or intermediate data objects

{qs2}: a framework for efficient serialization ... The qs2 format directly uses R serialization (via the R_Serialize/R_Unserialize C API) while improving underlying compression and disk IO patterns.

- .RData is terribly slow!
- .rds should now be much faster than before (serialized)
- .fst might be better in some cases but .qd/.qs is more generic (I think)

qs2

R-CMD-check **passing** CRAN **0.1.2** downloads **803/month** downloads **1274**

qs2: a framework for efficient serialization

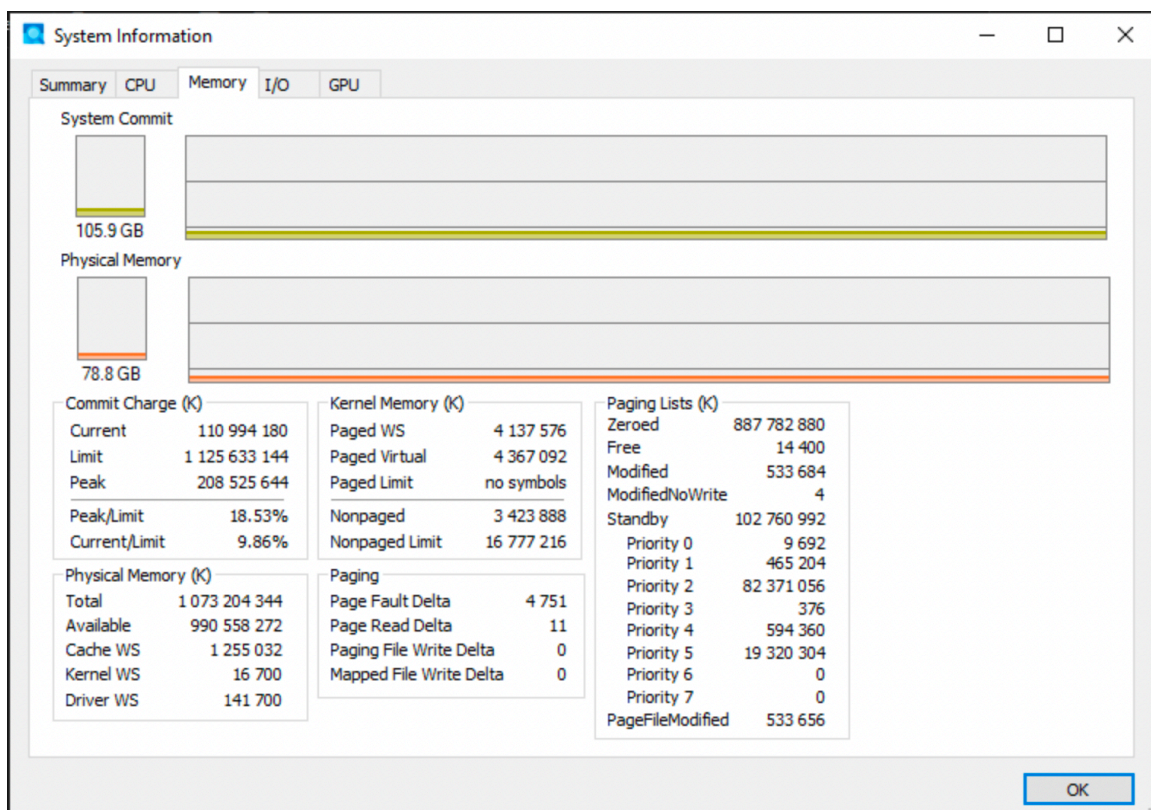
`qs2` is the successor to the `qs` package. The goal is to have reliable and fast performance for saving and loading objects in R.

The `qs2` format directly uses R serialization (via the `R_Serialize / R_Unserialize` C API) while improving underlying compression and disk IO patterns. If you are familiar with the `qs` package, the benefits and usage are the same.

```
qs_save(data, "myfile.qs2")
data <- qs_read("myfile.qs2")
```

Use the file extension `qs2` to distinguish it from the original `qs` package. It is not compatible with the original `qs` format.

Data management



- Let's say data fits into RAM
- base R and tidyverse works ... but might be terribly slow ...

- {data.table}! 😊

data.table 1.16.99
Introduction Vignettes News Benchmarks Presentations Articles Manual

data.table

data.table provides a high-performance version of [base R's data.frame](#) with syntax and feature enhancements for ease of use, convenience and programming speed.

Why data.table?

- concise syntax: fast to type, fast to read
- fast speed
- memory efficient
- careful API lifecycle management
- community
- feature rich

Links

- [View on CRAN](#)
- [Browse source code](#)
- [Report a bug](#)
- [CRAN-like website](#)
- [CRAN-like checks](#)
- [Community blog](#)

License

[MPL-2.0](#) | file [LICENSE](#)

Community

- [Contributing guide](#)

Citation

{data.table}

- Share basic properties with data frames (dt[i, j]; merge())
- Print method inspired by {tibble}
- Some old friends from {reshape2}: melt(), dcast()
- Has been around since 2006 (stable and well used, older than tidyverse)
- faster alternatives to base: fifelse(), fcase(), forder(), fread(), fcoalesce(), as.IDate(), %chin%
- C and GForce (some difficulties on Mac ...)
- integer based keys and indices (compare data bases)
- Reference semantics (no need to copy on modification)
- Very flexible and compact (but perhaps unusual) syntax

Example

```
library(data.table)
dt <- iris
setDT(dt)
setkey(iris)
iris
```

Key: <Sepal.Length, Sepal.Width, Petal.Length, Petal.Width, Species>

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
	<num>	<num>	<num>	<num>	<fctr>
1:	4.3	3.0	1.1	0.1	setosa
2:	4.4	2.9	1.4	0.2	setosa
3:	4.4	3.0	1.3	0.2	setosa

```

  4:      4.4      3.2      1.3      0.2  setosa
  5:      4.5      2.3      1.3      0.3  setosa
  ---
146:      7.7      2.6      6.9      2.3 virginica
147:      7.7      2.8      6.7      2.0 virginica
148:      7.7      3.0      6.1      2.3 virginica
149:      7.7      3.8      6.7      2.2 virginica
150:      7.9      3.8      6.4      2.0 virginica

```

```
dt[, .N, Species]
```

```

  Species      N
   <fctr> <int>
1:  setosa    50
2: versicolor 50
3:  virginica 50

```

```
dt[, date := as.IDate(Sys.Date()) + .GRP, Species][]
```

```

  Sepal.Length Sepal.Width Petal.Length Petal.Width Species      date
           <num>      <num>      <num>      <num>   <fctr>   <IDat>
1:          4.3          3.0          1.1          0.1  setosa 2026-03-15
2:          4.4          2.9          1.4          0.2  setosa 2026-03-15
3:          4.4          3.0          1.3          0.2  setosa 2026-03-15
4:          4.4          3.2          1.3          0.2  setosa 2026-03-15
5:          4.5          2.3          1.3          0.3  setosa 2026-03-15
  ---
146:         7.7          2.6          6.9          2.3 virginica 2026-03-17
147:         7.7          2.8          6.7          2.0 virginica 2026-03-17
148:         7.7          3.0          6.1          2.3 virginica 2026-03-17
149:         7.7          3.8          6.7          2.2 virginica 2026-03-17
150:         7.9          3.8          6.4          2.0 virginica 2026-03-17

```

```
dt[, names(.SD) := lapply(.SD, as.character), .SDcols = is.factor][]
```

```

  Sepal.Length Sepal.Width Petal.Length Petal.Width Species      date
           <num>      <num>      <num>      <num>   <char>   <IDat>
1:          4.3          3.0          1.1          0.1  setosa 2026-03-15
2:          4.4          2.9          1.4          0.2  setosa 2026-03-15
3:          4.4          3.0          1.3          0.2  setosa 2026-03-15
4:          4.4          3.2          1.3          0.2  setosa 2026-03-15
5:          4.5          2.3          1.3          0.3  setosa 2026-03-15
  ---

```

146:	7.7	2.6	6.9	2.3	virginica	2026-03-17
147:	7.7	2.8	6.7	2.0	virginica	2026-03-17
148:	7.7	3.0	6.1	2.3	virginica	2026-03-17
149:	7.7	3.8	6.7	2.2	virginica	2026-03-17
150:	7.9	3.8	6.4	2.0	virginica	2026-03-17

Tidyverse-like Data Manipulation built on *data.table*

- [tidytable](#): A tidy interface to *data.table* that is *rlang* compatible. Quite comprehensive implementation of *dplyr*, *tidyr* and *purrr* functions. Package uses a class *tidytable* that inherits from *data.table*. The `dt()` function makes *data.table* syntax pipeable (12 total dependencies).
- [dtplyr](#): A tidy interface to *data.table* built around lazy evaluation i.e. users need to call `as.data.table()`, `as.data.frame()` or `as_tibble()` to access the results. Lazy evaluation holds the potential of generating more performant *data.table* code (20 dependencies).
- [tidyfst](#): Tidy verbs for fast data manipulation. Covers *dplyr* and some *tidyr* functionality. Functions have `_dt` suffix and preserve *data.table* object. A [cheatsheet](#) is provided (7 dependencies).
- [tidyft](#): Tidy verbs for fast data operations by reference. Best for big data manipulation on out of memory data using facilities provided by *fst* (7 dependencies).
- [tidyfast](#): Fast tidying of data. Covers *tidyr* functionality, `dt_` prefix, preserves *data.table* object (2 dependencies).
- [maditr](#): Fast data aggregation, modification, and filtering with pipes and *data.table*. Minimal implementation with functions `let()` and `take()` for most common data manipulation tasks. Also provides Excel-like lookup functions (2 dependencies).
- [table.express](#) also builds *data.table* expressions from *dplyr* verbs, without executing them eagerly. Similar to *dtplyr* but less mature (17 dependencies).

Notes: These packages are wrappers around *data.table* and do not introduce own compiled code.

If it doesn't fit?

- Connect to parquet file.
- Use DuckDB to process the data (while still on disk)
- Use {dplyr} to translate its syntax into SQL code
- Process data on disk
- Only read the results into memory

Example

```
library(tidyverse)
db1 <-
  arrow::open_dataset("file1.parquet") |>
  arrow::to_duckdb() |>
  select(col1:col3) |>
  filter(...) |>
  pivot_wider(..)

db2 <-
  arrow::open_dataset("file2.parquet") |>
  arrow::to_duckdb() |>
  count(x, y) |>
  filter(...)

df <-
  db1 |>
  left_join(db2) |>
  collect() # Only the resulting part should be collected!
```

Pros 👍

- Lazy: nothing evaluated until it is actually needed (collected)
- Many basic functions are translated from R to SQL
- DuckDB is parallelized ({data.table} is mostly sequential)
- Familiar syntax from tidyverse

Cons 👎

- tidyverse -> SQL translation not perfect (non optimized)
 - might be relatively slow if intermediate steps are not cached and needs to be performed repeatedly
-

duckplyr 0.4.1

Reference

Changelog

Se



duckplyr

The goal of the duckplyr R package is to provide a drop-in replacement for [dplyr](#) that uses [DuckDB](#) as a backend for fast operation. DuckDB is an in-process OLAP database management system, dplyr is the grammar of data manipulation in the tidyverse.

duckplyr also defines a set of generics that provide a low-level implementer's interface for dplyr's high-level user interface.

Installation

Install duckplyr from CRAN with:

```
install.packages("duckplyr")
```

You can also install the development version of duckplyr from R-universe:

```
install.packages("duckplyr", repos = c("https://tidyverse.r-universe.dev", "
```

LINKS

[View on CRAN](#)
[Browse source code](#)
[Report a bug](#)

LICENSE

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COMMUNITY

[Contributing guide](#)

CITATION

[Citing duckplyr](#)

DEVELOPERS

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 posit
 Copyright holder, funder

DuckPlyr

- New package with API-calls to DuckDB (no standard SQL code in between ... I think)
- Drop-in-replacement of {dplyr}
- Automatically fail-safe to collect and use standard {dplyr} if needed
- Uptimized engine to not touch more data than necessary!
- Although very limited functionality so far (but wait and see!)

```

out <-
  palmerpenguins::penguins %>%
    # CAVEAT: factor columns are not supported yet
    mutate(across(where(is.factor), as.character)) %>%
    duckplyr::as_duckplyr_tibble() %>%
    mutate(bill_area = bill_length_mm * bill_depth_mm) %>%
    summarize(.by = c(species, sex), mean_bill_area = mean(bill_area)) %>%
    filter(species != "Gentoo")

out %>%
  explain()

```

```

#> [-----]
#> ORDER_BY
#> [-----]
#> dataframe_42_42
#> 42.__row_number ASC
#> [-----]
#> |
#> [-----]
#> FILTER
#> [-----]
#> "r_base::!="(species,
#> 'Gentoo')
#>
#> ~34 Rows
#> [-----]
#> |
#> [-----]
#> PROJECTION
#> [-----]
#> #0
#> #1
#> #2
#> #3
#>
#> ~172 Rows
#> [-----]
#> |
#> [-----]
#> STREAMING_WINDOW
#> [-----]
#> Projections:
#> ROW_NUMBER() OVER (
#>
#> [-----]
#> |
#> [-----]
#> ORDER_BY
#> [-----]
#> dataframe_42_42
#> 42.__row_number ASC
#> [-----]
#> |
#> [-----]
#> HASH_GROUP_BY
#> [-----]
#> Groups:
#> #0
#> #1
#>
#> Aggregates:
#> min(#2)
#> mean(#3)
#>
#> ~172 Rows

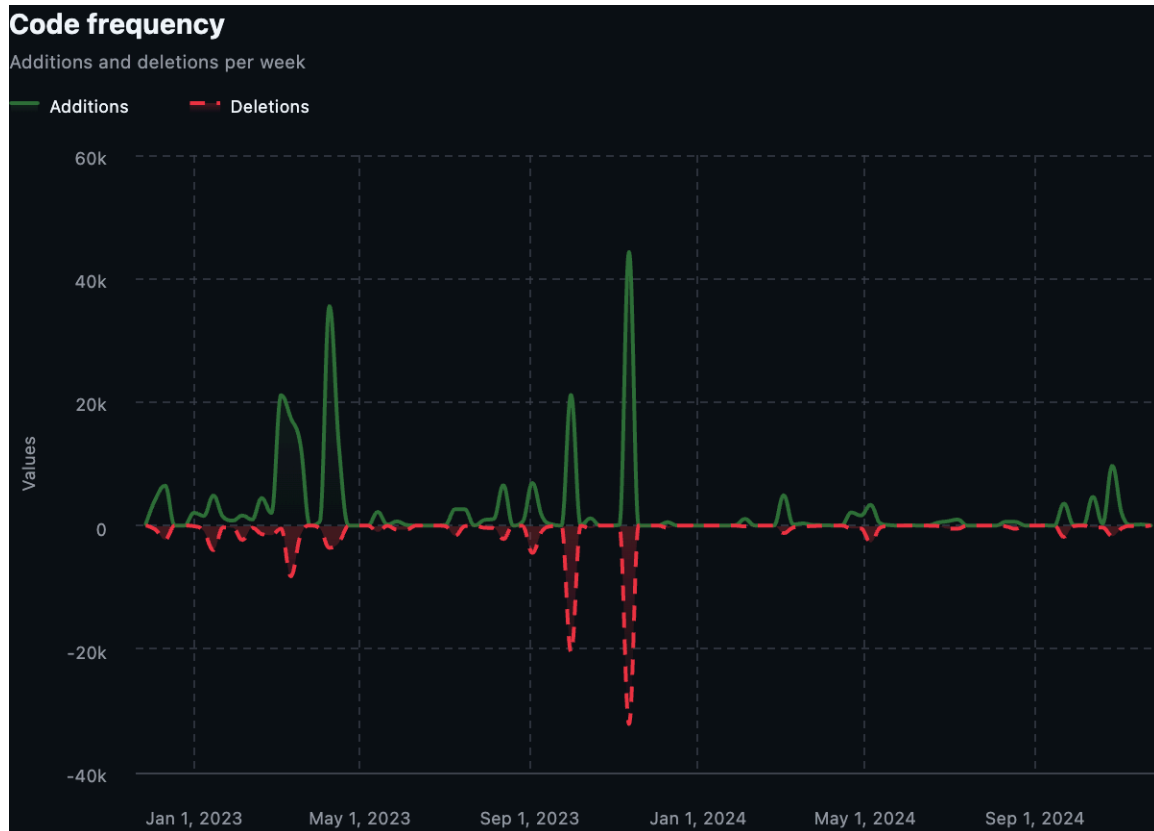
```

```

#> ┌──────────┴──────────┐
#> ┌──────────┴──────────┐
#> PROJECTION
#> ───────────
#> species
#> sex
#> __row_number
#> bill_area
#>
#> ~344 Rows
#> ┌──────────┴──────────┐
#> ┌──────────┴──────────┐
#> PROJECTION
#> ───────────
#> #0
#> #1
#> #2
#> #3
#>
#> ~344 Rows
#> ┌──────────┴──────────┐
#> ┌──────────┴──────────┐
#> STREAMING_WINDOW
#> ───────────
#> Projections:
#> ROW_NUMBER() OVER ( )
#> ┌──────────┴──────────┐
#> ┌──────────┴──────────┐
#> PROJECTION
#> ───────────
#> species
#> sex
#> bill_area
#>
#> ~344 Rows
#> ┌──────────┴──────────┐
#> ┌──────────┴──────────┐
#> R_DATAFRAME_SCAN
#> ───────────
#> data.frame
#>
#> Projections:
#> species
#> bill_length_mm
#> bill_depth_mm
#> sex
#>

```

```
#> | ~344 Rows |  
#> |-----|
```

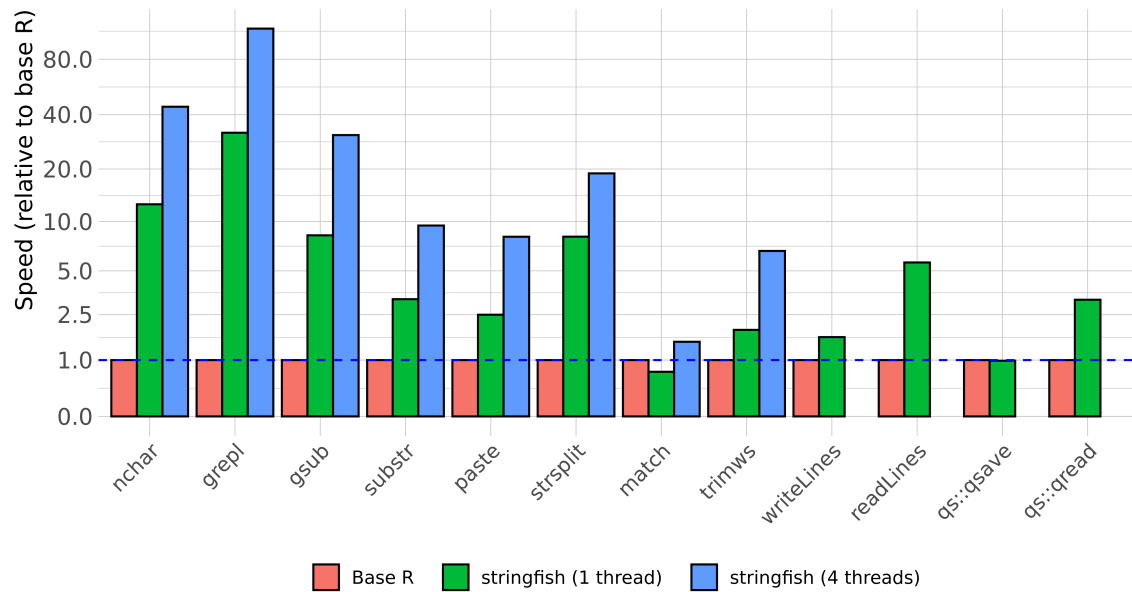


Dates

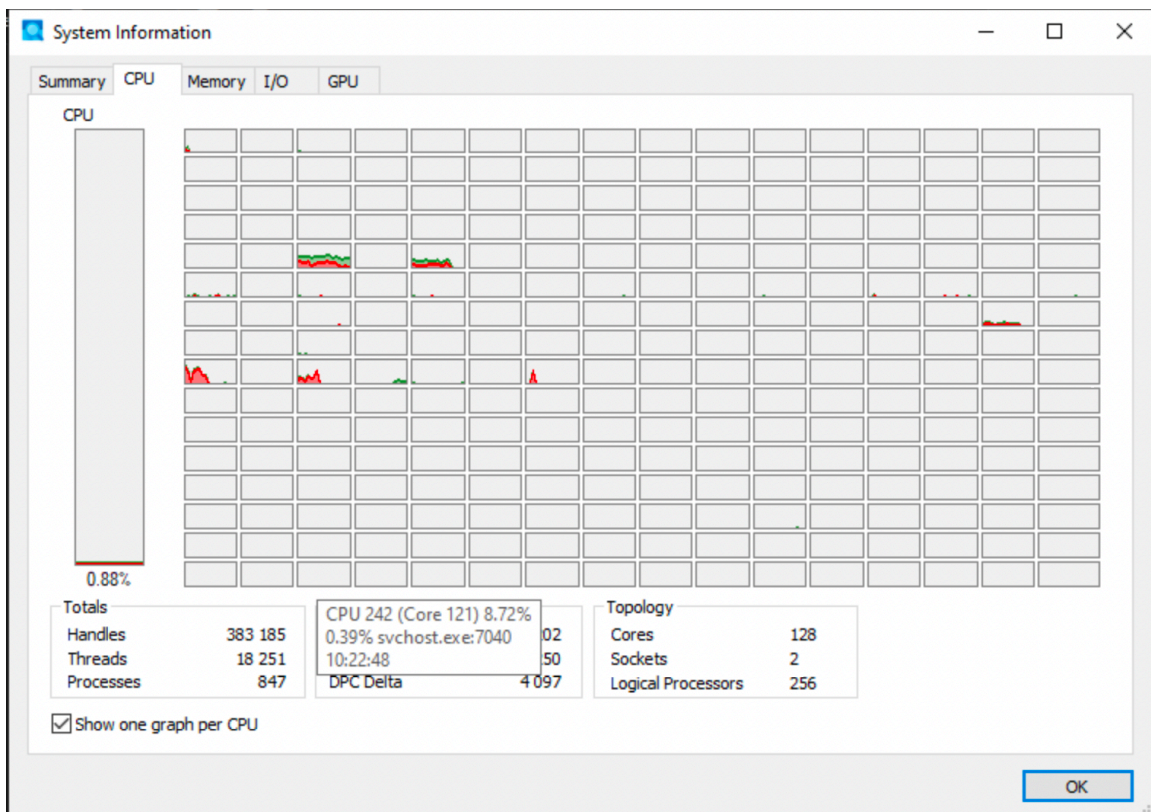
- Use `as.IDate()` instead of `as.Date()` if using `{data.table}`. Integer based.
- Use `{anytime}` (or maybe `{fasttime}`) if conversion is needed (ALTREP)

String manipulation

- Character manipulation in base R can be quite slow (does not use ALTREP and only single core by default)
 - For example regular expressions (`grep()`, `gsub()` etc)
- `{stringfish}` use multithreads and ALTREP - much faster!



Paralellization



Paralellization

- R is single threaded by default
- Some packages and functions may use multithreading by default
- Some functions might have arguments such as `nthreads`, `ncores` etc
- To handle multicore/thread computations can be (very) difficult (I've heard)
- Easy if data can be divided and conquered (split-apply-combine)
- `{futureverse}` is my favorite (Henrik Bengtsson)
 - `{furrr}` is a drop-in replacement for `{purrr}`

`{furrr}`

```
# How many cores do we have?  
parallel::detectCores()
```

```
[1] 12
```

```
library(furrr)
```

```
Loading required package: future
```

```
# Save at least one for OS etc  
plan(multisession, workers = parallel::detectCores() - 1)  
  
1:10 |>  
  future_map(rnorm, n = 10, .options = furrr_options(seed = 123)) |>  
  future_map_dbl(mean)
```

```
[1] 1.180279 2.140442 2.909823 3.692207 5.058100 6.653926 7.065630 7.960713  
[9] 9.105674 9.766827
```

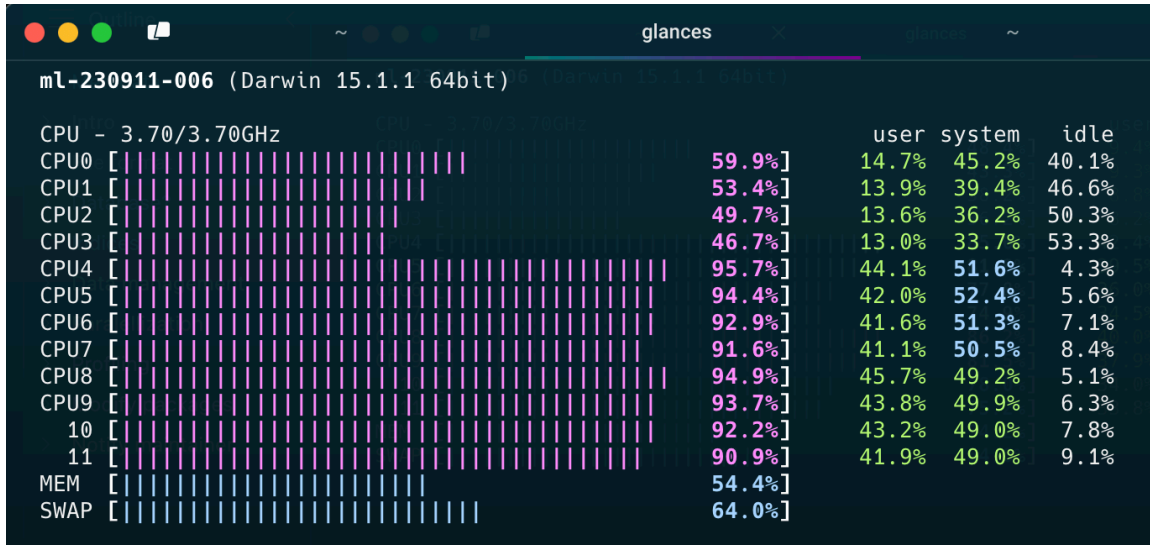
`{mirai}`

- Suposed to be up to 1,000 times “faster” compared to `{furrr}`(according to some metric ...)
- should not block the main process while executing (haven't tried)

Designed for simplicity, a ‘mirai’ evaluates an R expression asynchronously in a parallel process, locally or distributed over the network, with the result automatically available upon completion.

```
library(mirai)  
daemons(11, dispatcher = "thread") # "thread" still experimental
```

```
# data.table with 1 billion rows:
dt <- data.table(x = seq_len(1e9), y = rnorm(1e9))
# row sums (row wise )
mirai_map(dt, sum)[.flat]
```



Note

- Data must be copied for each parallel worker
- If all data is needed for each worker, it will be multiplied in memory
 - This might not be possible for big data
 - It also takes time
- Hence, there is no guarantee that the over all time will decrease
- Often a tradeoff between no of workers/threads/deamons/cores and efficiency etc
- “Standard” computations in {data.table} is optimized for single core
- Message handling and progress bar etc is complicated

Caching

- I use {ProjectTemplate} to organize R projects into different steps
- It has an inbuilt cache mechanism through ProjectTemplate::cache()
- I have my own cache-functions in my .Rprofile

```
# OBS kollar inte att det verkligen finns cache-folder etc
cache_dt <- function(x, name = deparse(substitute(x)), nthreads = 5) {
  qs2::qd_save(x, glue::glue("cache/{name}.qd"), nthreads = nthreads)
  invisible(x)
}
```

```

decache_dt <- function(name, nthreads = 5) {
  x <- qs2::qd_read(glue::glue("cache/{name}.qd"), TRUE, nthreads = nthreads)
  if (data.table::is.data.table(x)) {
    data.table::setDT(x)
  }
  assign(name, x, envir = .GlobalEnv)
  invisible(x)
}

```

Intermediate caching

- For long running function calls (hours, days)
- Divide task into smaller pieces and execute each part in parallel if feasible
- Use wrapper function with some index/grouping variable to indicate which part is currently processed
- Cache intermediate object as soon as possible (with name based on the grouping variable)
- If process is disrupted and must be restarted, parts which are already executed are just retrieved from cache

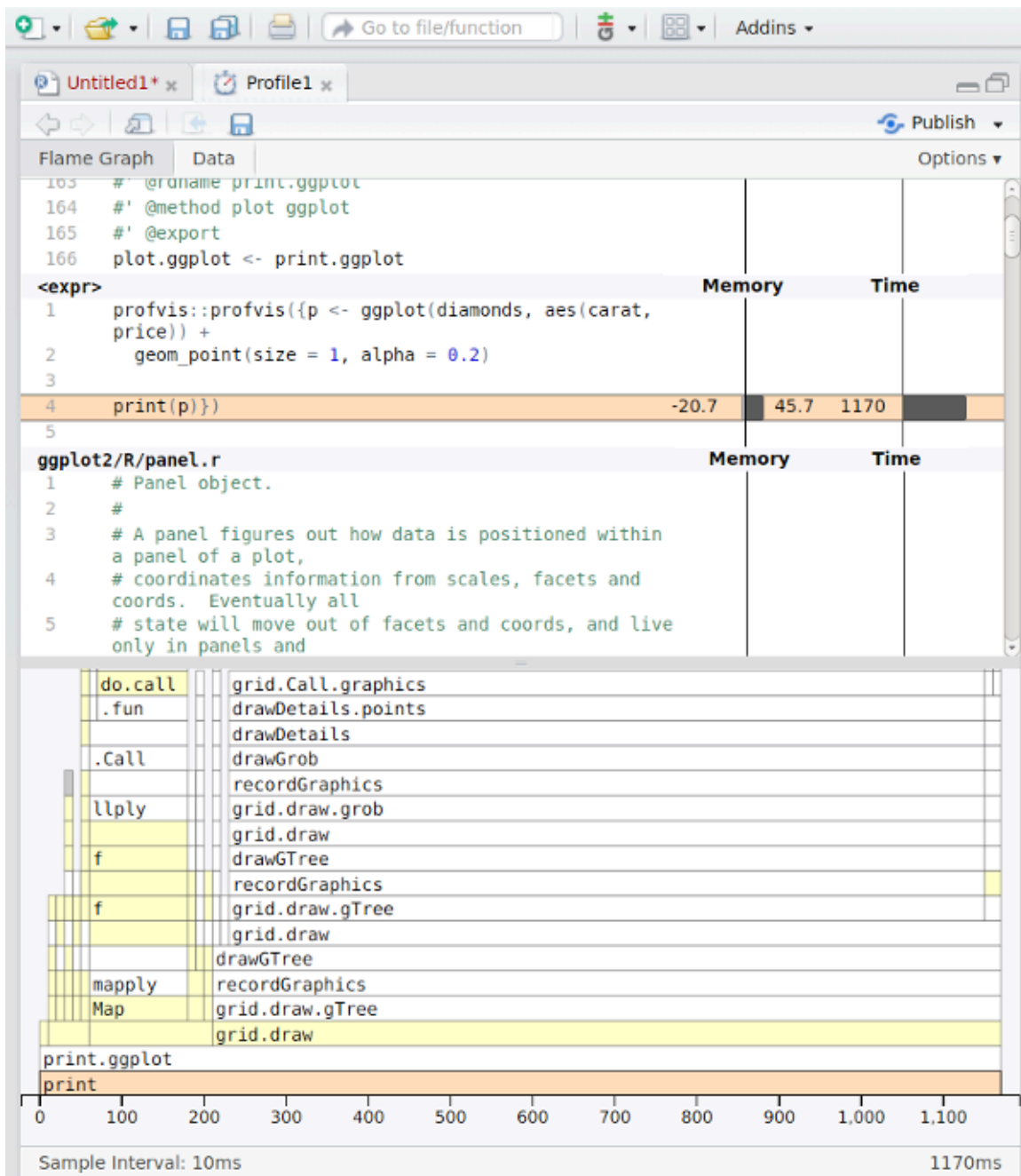
```

cache_wrapper <- function(dt_part, name, ...) {
  if (fs::file_exists(glue::glue("cache/{name}.qd"))) {
    decache_dt(name)
  } else {
    x <- do_what_should_be_done()
    cache_dt(x, name, ...)
  }
  x
}

```

Profiling

- First version of R script might be inefficient
 - Optimization is difficult
 - Might be suboptimal if made ad hoc
 - {profvis} visualise time and memory usage for each step
 - improve the most important step first
 - For example change {base} and {tidyverse} code to {data.table}
 - set keys
 - Efficient handling of dates and strings
 - iterate
 - Integrated in RStudio (but not yet Positron)
-



Modify packages

Modify packages

- Functions from packages might be inefficient
- Profile those as well
- Clone package from GitHub if available or download source code from CRAN
- Improve

- Just load the modified function in global environment
 - might need to modify internal calls to `:::-` accessed functions
- Or rebuild/install the package if more convenient

! Notify maintainer

If you find ways to improve a package, the maintainer might be eager to hear! Suggest pull request or open GitHub issue!

C-code changes

- If the package use C-code it is probably already very fast
- If not, and you can fix it, the package must be re-compiled
- OS dependent
- Might be tricky if using restricted environments (TRE?)
- Can still be done by rhub (might need to change maintainer-e-mail temporarily)

rhub
2.0.0
Reference
Articles
News

rhub

R-hub v2

R-hub v2 uses GitHub Actions to run `R CMD check` and similar package checks. The `rhub` package helps you set up R-hub v2 for your R package, and start running checks.

- Installation
- Usage
 - Requirements
 - Private repositories
 - Setup
 - Run checks
- The R Consortium runners
 - Limitations of the R Consortium runners
- Code of Conduct
- License

Installation

Install `rhub` from CRAN:

```
pak::pkg_install("rhub")
```

Links

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Citation

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fastverse 0.3.4
DOCUMENTATION
VIGNETTES
NEWS
R-UNIVERSE
BLOG

Search for

fastverse

The *fastverse* is a suite of complementary high-performance packages for statistical computing and data manipulation in R. Developed independently by various people, *fastverse* packages jointly contribute to the objectives of:

- Speeding up R through heavy use of compiled code (C, C++, Fortran)
- Enabling more complex statistical and data manipulation operations in R
- Reducing the number of dependencies required for advanced computing in R

The **fastverse** package is a meta-package providing utilities for easy installation, loading and management of these packages. It is an extensible framework that allows users to (permanently) add or remove packages to create a 'verse' of packages suiting their general needs, or even create separate 'verses' of their own.

fastverse packages are jointly attached with `library(fastverse)`, and several functions starting with `fastverse_` help manage dependencies, detect namespace conflicts, add/remove packages from the *fastverse* and update packages. The [vignette](#) provides a concise overview of the package.



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Bibliography